

Unit 20

POLYMERISATION

14 items · 27 marks

Syllabus sections in this unit

20.1 Addition polymerisation

Unit 20: Polymerisation

1. M/J 26/P21/Q5(f)

Context

- 5 Figure 5.1 describes a sequence of reactions starting with ethene to produce poly(chloroethene), PVC.

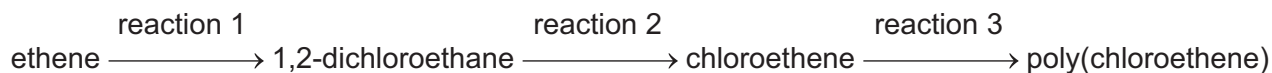


Figure 5.1

- (f) Draw the repeat unit of poly(chloroethene).

[1]

2. M/J 26/P24/Q5(b)(i)-(ii)

Context

- 5 Long-chain alkanes from crude oil can be cracked to make alkenes.

Context: part (b) (introduction)

- (b) Propene is an alkene that is made by cracking. Alkenes are used in addition polymerisation.

- (i) Draw **one** repeat unit of the addition polymer formed from propene.

[1]

- (ii) The disposal of poly(alkenes) by combustion produces harmful products.

State **one** other disadvantage associated with the disposal of poly(alkenes) such as poly(propene).

[1]

3. O/N 25/P21/Q4(b)(i)

Context

- 4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

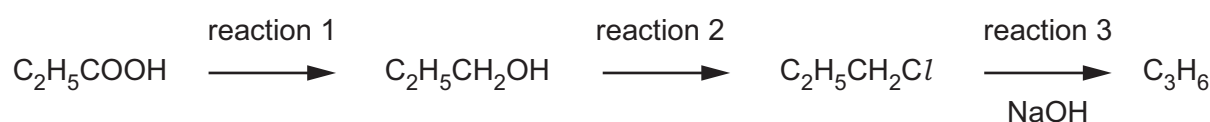


Fig. 4.1

Context: part (b) (introduction)

- (b) Under suitable conditions, propene polymerises to form poly(propene).

Poly(propene) exhibits stereoisomerism.

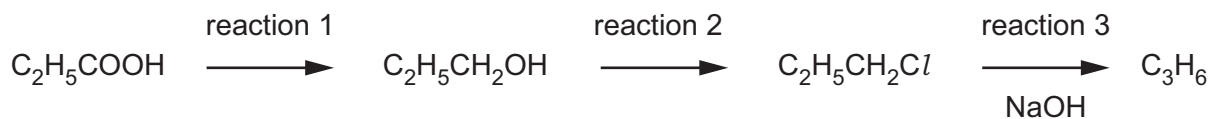
- (i) Identify the type of polymerisation that forms poly(propene) from propene.

[1]

4. O/N 25/P21/Q4(b)(iii)-(iv)

Context

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

Context: part (b) (introduction)

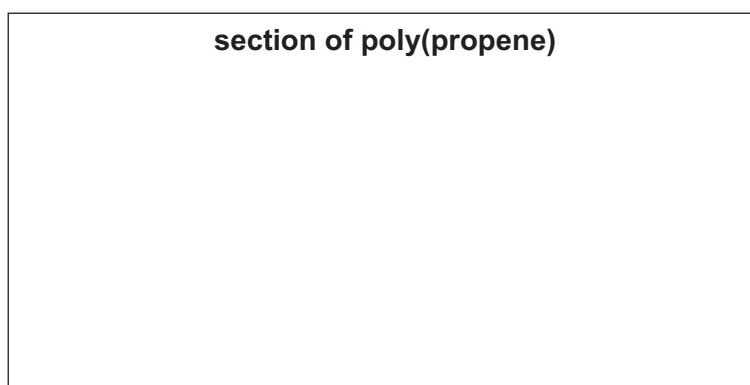
(b) Under suitable conditions, propene polymerises to form poly(propene).

Poly(propene) exhibits stereoisomerism.

(iii) Draw a section of poly(propene), showing **two** repeat units.

Use your diagram to identify the type of stereoisomerism shown by poly(propene).

Explain your answer.



[3]

(iv) State **two** difficulties associated with the disposal of poly(propene).

1

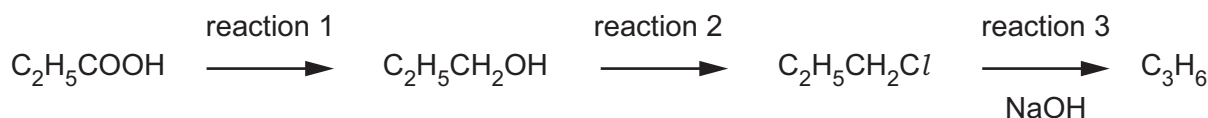
2

[2]

5. O/N 25/P23/Q4(b)(i)

Context

4 Fig. 4.1 shows a possible synthesis of propene, C_3H_6 .

**Fig. 4.1**

Context: part (b) (introduction)

(b) Under suitable conditions, propene polymerises to form poly(propene).

Poly(propene) exhibits stereoisomerism.

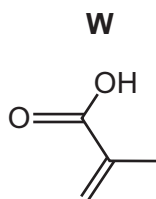
(i) Identify the type of polymerisation that forms poly(propene) from propene.

[1]

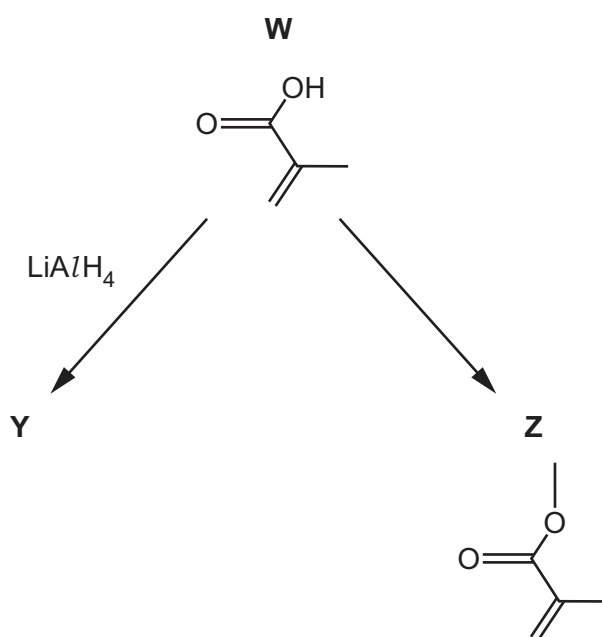
6. O/N 25/P23/Q4(b)(iii)-(iv)

7. M/J 25/P23/Q5(b)(vi)

Context: part (a) (introduction)

5 (a) W is a colourless liquid.**Fig. 5.1**

Context: part (b) (introduction)

(b) Fig. 5.2 shows two reactions of **W** to produce organic compounds **Y** and **Z**.**Fig. 5.2****(vi)** **Z** is used to produce addition polymer **Q**.Draw the repeat unit of polymer **Q**.

[1]

8. M/J 25/P23/Q5(c)**(c)** Poly(ethene) is an addition polymer made from ethene.Explain why ethene reacts with electrophiles but poly(ethene) does **not**.

[1]

9. MAR 24/P22/Q2(d)(iii)

Context

2 Chlorine, Cl_2 , reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

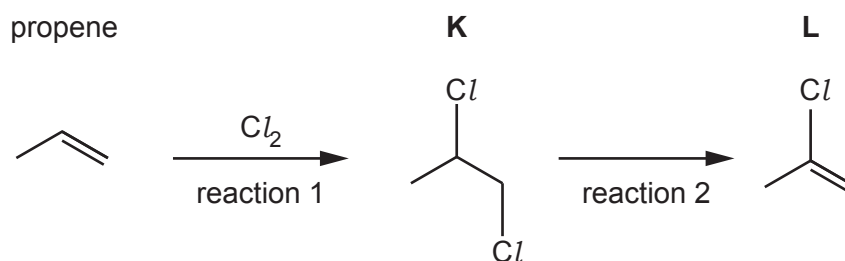
Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

Context: part (d) (introduction)

(d) Cl_2 reacts readily with propene to form **K**, 1,2-dichloropropane.

K can be used to form **L**.

**Fig. 2.1**

(iii) Draw **one** repeat unit of the addition polymer that forms from **L**.

[1]

10. M/J 23/P23/Q6(a)

Context

6 Compound **W**, $\text{CH}_2=\text{CHCN}$, is used to make an addition polymer which is present in carbon fibres.

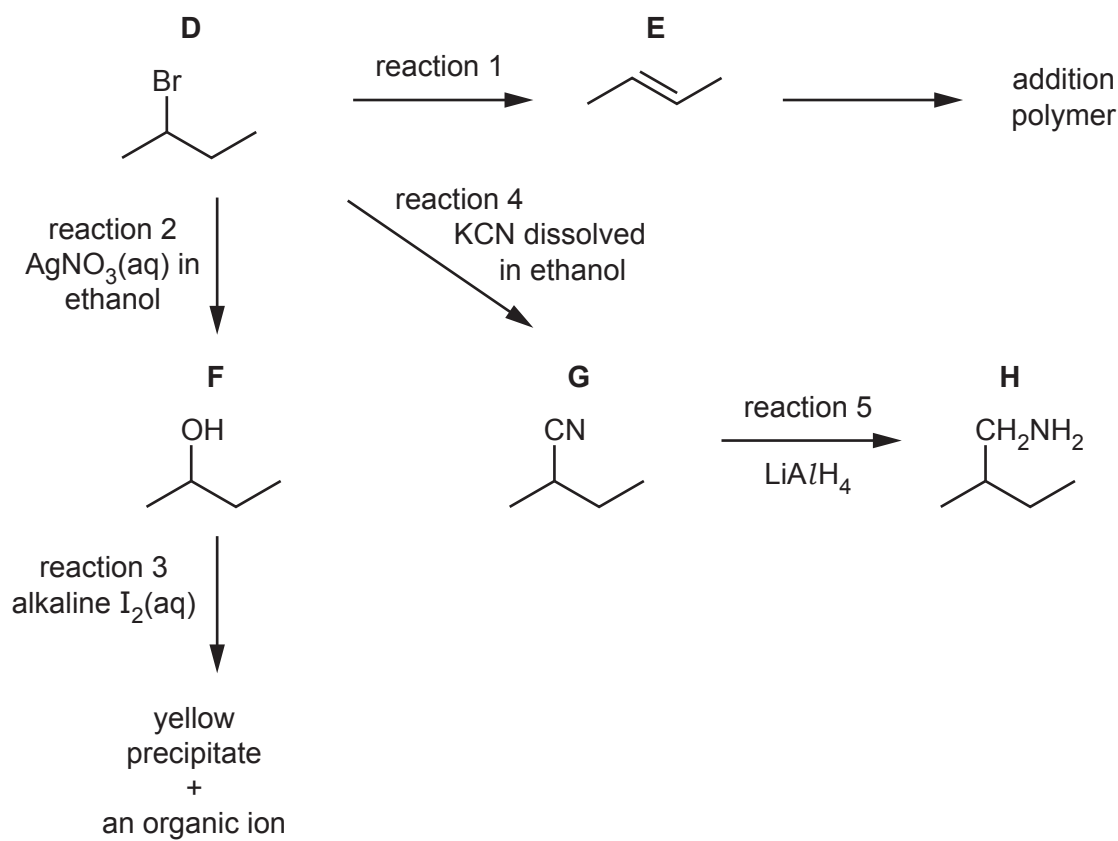
(a) Draw **one** repeat unit of the addition polymer of **W**.

[1]

11. MAR 23/P22/Q4(a)(ii)

Context

4 Fig. 4.1 shows some reactions of compound **D**, 2-bromobutane.

**Fig. 4.1**

(ii) Draw the structure of **one** repeat unit of the addition polymer that forms from **E**.

[1]

12. O/N 22/P22/Q3(b)

Context

3 Organic compounds can be distinguished using chemical tests and analytical techniques.

(b) **D2** forms polymer **Z** when heated gently.

(i) Identify the type of polymer that forms from **D2**.

[1]

(ii) Draw one repeat unit of polymer **Z**.

[2]

13. M/J 22/P22/Q4(c)

(c) **X** is an addition polymer.

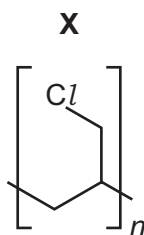


Fig. 4.3

(i) Draw the monomer of **X**.

[1]

(ii) Suggest **one** reason why the disposal of items made from **X** is difficult.

[1]

14. MAR 22/P22/Q4(b)(ii)

Context

4 Compounds **J** and **K** are found in plant oils.

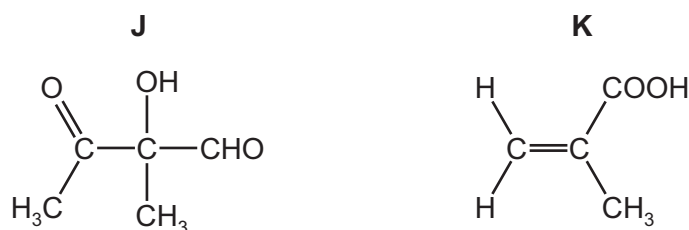


Fig. 4.1

Context: part (b) (introduction)

(b) **K** is used to make the addition polymer Perspex®. A synthesis of Perspex® is shown in Fig. 4.2.

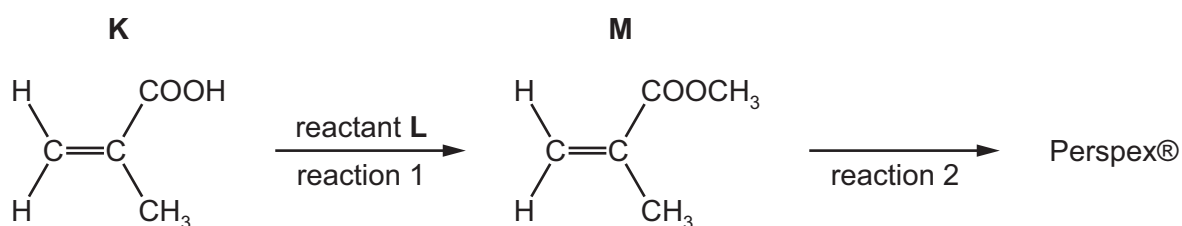


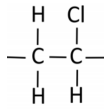
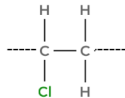
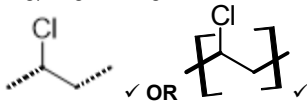
Fig. 4.2

(ii) Draw **one** repeat unit of the addition polymer Perspex®.

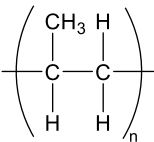
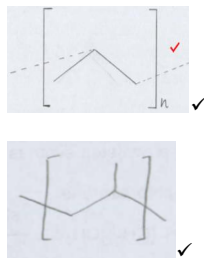
[2]

Answers Section

1. M/J 26/P21/Q5(f)

5(f) OR  OR 		1 I brackets (any) I n A Cl on either C atom A dangling bonds / bracketed repeat unit If skeletal then dangling bonds are different (e.g. dashed, v. long) or go through brackets 
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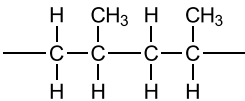
2. M/J 26/P24/Q5(b)(i)-(ii)

5(b)(i) 		1 I brackets I n A skeletal structure <u>with</u> hanging bonds / identifiable continuation bonds (i.e. dashed lines): 
5(b)(ii)	non-biodegradable	1 I produces harmful products / toxic gas I other environmental effects I disposal / cost A non-photodegradable A unreactive due to strong / non-polar bonds A phonetical spelling A BOD difficult / hard to decompose / does not decompose / does not break up / down (easily)

3. O/N 25/P21/Q4(b)(i)

4(b)(i)	addition	1
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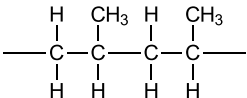
4. O/N 25/P21/Q4(b)(iii)-(iv)

4(b)(iii) M1 M2 optical isomerism M3 the methylated carbon has four different groups attached		3
4(b)(iv)	M1 non-biodegradability M2 harmful combustion products	2

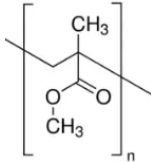
5. O/N 25/P23/Q4(b)(i)

4(b)(i)	addition	1
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6. O/N 25/P23/Q4(b)(iii)-(iv)

4(b)(iii)	 M1 M2 optical isomerism M3 the methylated carbon has four different groups attached	3
4(b)(iv)	M1 non-biodegradability M2 harmful combustion products	2

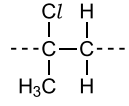
7. M/J 25/P23/Q5(b)(vi)

5(b)(vi)		1
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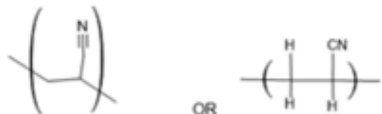
8. M/J 25/P23/Q5(c)

5(c)	ethene pi bond is a region of (high) electron density (above and below the C-C (sigma) bond)	1
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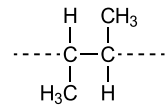
9. MAR 24/P22/Q2(d)(iii)

2(d)(iii)		1

10. M/J 23/P23/Q6(a)

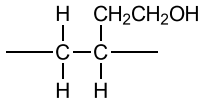
6(a)		1
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11. MAR 23/P22/Q4(a)(ii)

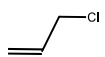
4(a)(ii)		1
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12. O/N 22/P22/Q3(b)

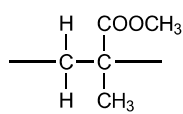
3(b)(i)	addition	1
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3(b)(ii)	 M1 correct carbon backbone of one repeat unit including 'dangling' bonds	1
	M2 rest of structure correct	1

13. M/J 22/P22/Q4(c)

4(c)(i)		1
4(c)(ii)	non-biodegradable OR poisonous / harmful <i>HCl</i> made when burnt	1

14. MAR 22/P22/Q4(b)(ii)

4(b)(ii)	 carbon backbone with 'dangling' bonds [1] rest of structure correct [1]	2
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